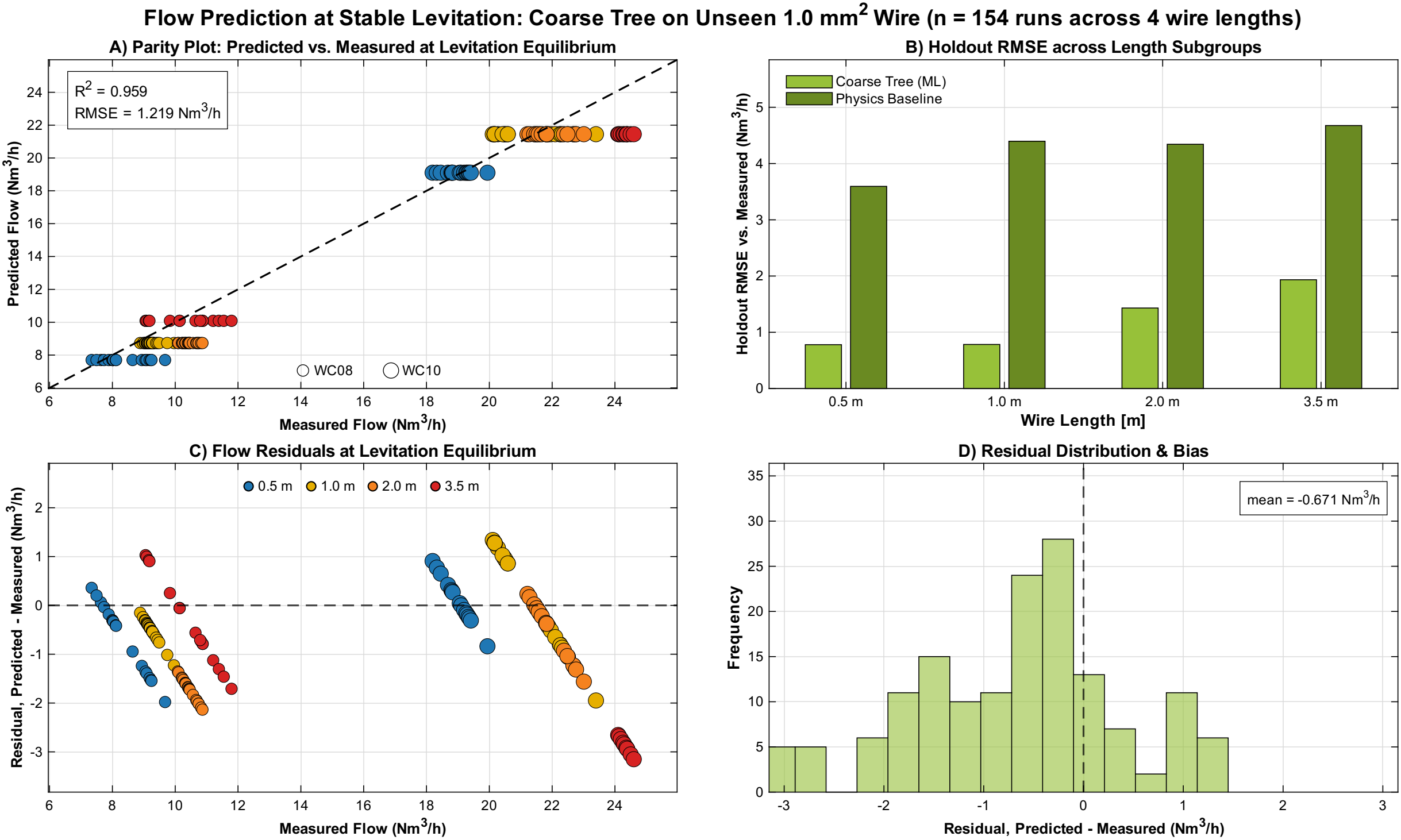

Flow Prediction at Stable Levitation: Coarse Tree on Unseen 1.0 mm² Wire (n = 154 runs across 4 wire lengths)
A) Parity Plot: Predicted vs. Measured at Levitation Equilibrium
R² = 0.959
RMSE = 1.219 Nm³/h
WC08
WC10
Predicted Flow (Nm³/h)
Measured Flow (Nm³/h)
B) Holdout RMSE across Length Subgroups
Coarse Tree (ML)
Physics Baseline
Holdout RMSE vs. Measured (Nm³/h)
0.5 m
1.0 m
2.0 m
3.5 m
Wire Length [m]
C) Flow Residuals at Levitation Equilibrium
0.5 m
1.0 m
2.0 m
3.5 m
Residual, Predicted - Measured (Nm³/h)
Measured Flow (Nm³/h)
D) Residual Distribution & Bias
mean = -0.671 Nm³/h
Frequency
Residual, Predicted - Measured (Nm³/h)